

Frontier Is Not Fate

What does a better AI model actually change?

Follow it into a workflow. Then ask what else has to happen before the improvement becomes a useful outcome.

One supported starting point: in the ECB's Q4 2025 survey, **33%** of responding firms reported very infrequent or experimental AI use; **7%** reported significant use.

Self-reported intensity among weighted SAFE respondents. This tells us how firms report using AI; it does not measure a productivity gain. [Source and limits](#)

Draft personal copy · for Jinhua's final edit

When a country gets access to better AI, what does it actually get? I kept returning to that question after conversations in Washington and my experience at the World AI Conference in Shanghai.

I wanted to follow the technology into a workplace or a laboratory: who uses it, what changes in the work, and who gets something useful from the change. This edition starts with a smaller question I can investigate—what happened when particular people and machines received particular AI tools?

This edition follows AI access into support work, software tasks and a plasma-control experiment. Its narrower answer concerns changes in those settings. It offers a way to investigate a model upgrade, rather than an estimate of the upgrade's marginal effect.

What has actually changed in work?

Start by separating reported use from measured work. A firm can tick "yes" while still experimenting. Surveys can tell us about reach, intensity or breadth; none of those answers alone

establishes a completed outcome.

Observed: reported use has several meanings.

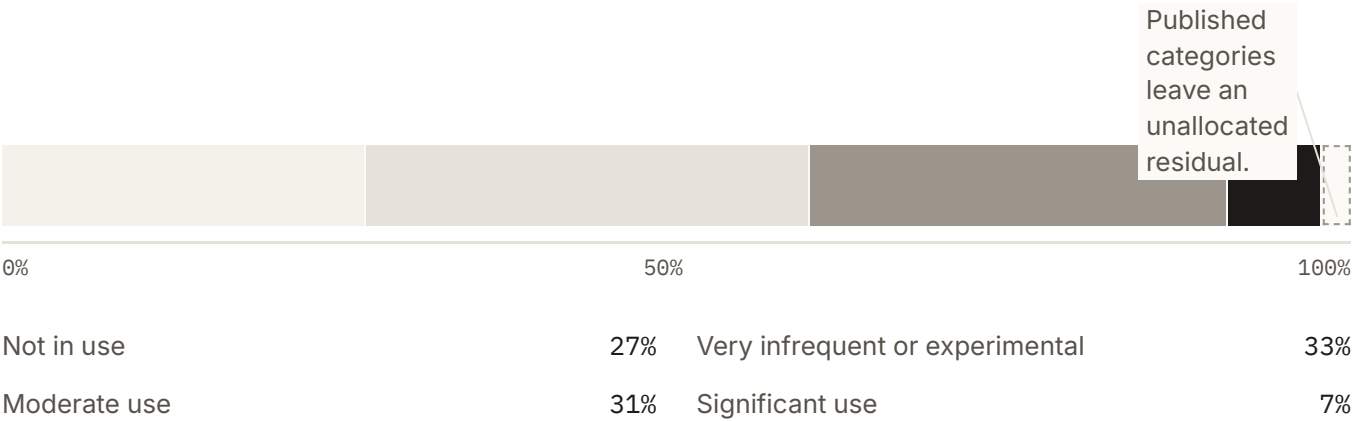
Percent of firms within each source’s own universe. ECB intensity, Census function breadth and Eurostat firm-size adoption stay separate.

Each panel retains its own axis, denominator, and survey universe. Values are source-specific and must not be pooled or ranked.

PANEL A · ECB SAFE · EURO AREA

Most reported use is experimental or moderate.

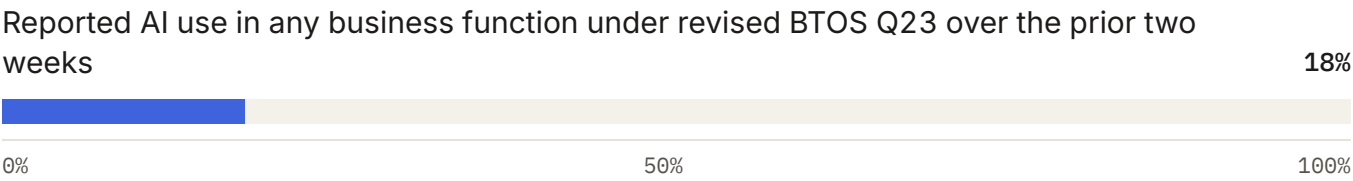
AXIS 0-100% · WEIGHTED SAFE RESPONDENTS



ECB SAFE Q4 2025 – source and caveat · denominator: All firms responding to SAFE QA1, weighted

Reach and functional breadth answer different questions.

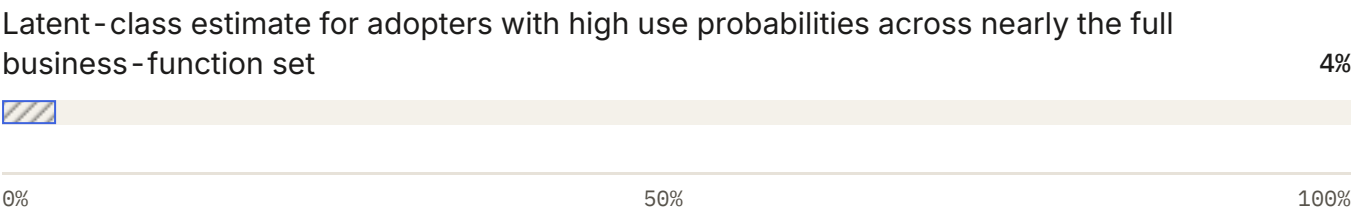
ALL EMPLOYER BUSINESSES · AXIS 0-100%



Q24 FUNCTIONAL USERS ONLY · AXIS 0-100%



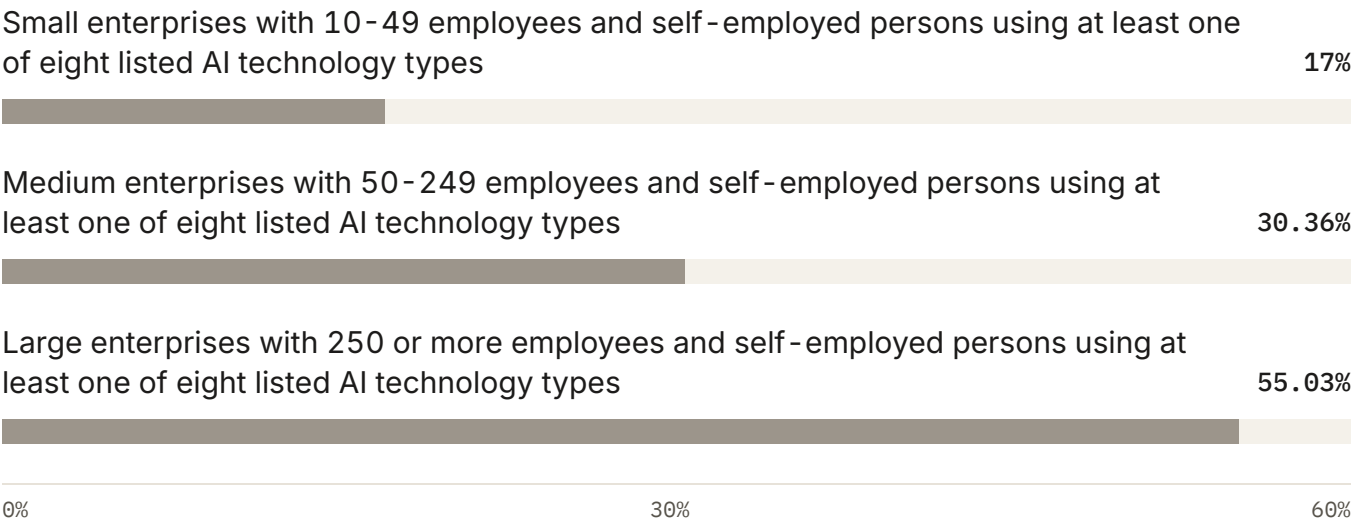
MODEL ESTIMATE



Census CES Working Paper 26-25 – [source and caveat](#) · Q23 denominator: All in-scope employer businesses responding to Q23, firm-weighted · Q24 denominator: Firms reporting AI use in at least one of 15 Q24 business functions, firm-weighted

Adoption is uneven across firm size.

WITHIN EACH SIZE UNIVERSE · AXIS 0-60%



Diffusion context only. This panel does not measure depth of use.

Eurostat 2025 – source and caveat · universe: Enterprises with 10 or more employees and self-employed persons in covered NACE Rev. 2 sectors; micro-enterprises are optional and outside the harmonized core

A different firm universe gives context only.

The NBS Fifth National Economic Census covers a different enterprise universe. It cannot supply a comparable all-firm adoption rate or a national ranking here.

The contextual observation is retained in the research record, unplotted. Load-bearing translated wording awaits native-language human review.

NBS – scope and translation restriction

ECB SAFE Q4 2025; Census CES Working Paper 26-25; Eurostat 2025 enterprise AI use. NBS context retained, unplotted. **OBSERVED** DEFINITIONS DIFFER

Evidence and caveats

All firms responding to SAFE QA1, weighted; 5,067-firm Q4 ECB-round sample across 12 euro-area countries with sector exclusions. The four whole-number categories total 98%; the residual may reflect don't-know responses and rounding, so do not derive 73% any use.

Denominator is all firms responding to SAFE QA1, weighted, in the 5,067-firm Q4 ECB-round private-sector sample across 12 euro-area countries; eight smallest euro-area countries and documented sectors are excluded. This is a source-specific category interpretation; do not infer a harmonized deep-use threshold or calculate a 73% any-use share from the 27% no-use value.

Firm-weighted denominator is businesses responding to Q23 within the nonfarm employer-business BTOS universe and documented exclusions. The paper headline rounds to 18%; Table C.7 reports 17.9%. Revised wording began 2025-11-17.

Denominator is firms reporting AI use in at least one Q24 function within the nonfarm U.S. employer-business universe covering single- and multi-location businesses in the United States, District of Columbia, and Puerto Rico, subject to documented industry exclusions; it is not the Q23 prior-two-week adopter group. The 4% is model-derived, does not mean use in every function, and does not form a partition with 57%.

The study universe is nonfarm U.S. employer businesses, single- and multi-location, in the United States, District of Columbia, and Puerto Rico, subject to documented BTOS industry exclusions; the analysis uses supplement respondents. The paper states that regressions have no causal interpretation and that CES papers do not imply Census Bureau endorsement.

Denominators are enterprises with 10+ employees and self-employed persons in covered NACE activities, with size-specific subgroups; micro-enterprises are outside the harmonized core. This is adoption by firm size, not use depth.

Observation verification: 2026-08-21. Edition date is not a new measurement date.

Sources and observation periods

The useful result is that a yes/no adoption count hides different reported practices. One plausible explanation is that firms are still finding workable uses; another is that intensive use is unnecessary or too costly for some tasks. The survey does not decide between them. Higher use is not automatically better, and non-adoption is not automatically failure.

What the survey can help us ask

These categories describe reported behavior, not a maturity ladder. Experimental use could precede integration, remain a useful occasional tool, or end after an unsuccessful trial. A stronger model might make a previously unsuitable task worth attempting; prices, organizational needs or an effective existing process could still explain a firm's choice. Outcome evidence is needed to judge that choice.

What changed when people had access?

The next studies measure completed work. Read each on its own terms before asking why the results differ.

Observed: different work, different measured effects.

Separate outcomes and populations; no combined AI-productivity average. Study-specific sources and exact-use review status below.

Access is the intervention here. These studies compare access to an AI assistant with no access in particular settings. They do not isolate an upgrade from a weaker model to a stronger one. Differences between the findings cannot be attributed to workflow design alone: tools, people, tasks, dates and estimators also differ.

Customer support

More resolved issues per hour

Population: 5,172 agents at one business-software firm and subcontractors; most worked in the Philippines serving US customers.

Tools / study vintage: GPT-3-based assistant; Main rollout: fall 2020 / winter 2021.

Resolutions/hour index · reference = 100



Transformation: $100 \times (1 + 15/100) = 115$. These are normalized reported relative estimates, not raw observed group means. Each panel has a different unit; the indices do not create a common productivity metric.

In one customer-support deployment, access to a GPT-3-based assistant increased issues resolved per hour by about 15%.

One firm and an older assistant. Comparable counterfactual trends are an identification assumption. No measured national productivity, wage or welfare effect.

The useful outcome is more resolved issues in an hour of support work. Who keeps the gain depends on staffing, compensation, service commitments and ownership—quantities this percentage does not allocate.

How to read this result

A plausible mechanism · source-supported account

The paper finds larger gains among less experienced and lower-skilled agents and evidence consistent with the assistant spreading effective practices. A suggestion available during a customer conversation can make useful know-how easier to apply.

The strongest alternative · analyst hypothesis

The rollout could coincide with other changes affecting treated agents. Even if the average access effect is credible, skill patterns do not uniquely identify knowledge transfer as its cause.

When a stronger model could matter · analyst hypothesis

An upgrade could matter more if it resolves difficult cases reliably enough to change which work an agent can finish. It could matter less if existing suggestions already meet the need, or if checking additional suggestions consumes the gain. That requires an upgrade comparison.

Design, uncertainty and quality

Comparison

Assistant access versus no access in a staggered rollout

Design

Staggered rollout; difference-in-differences with agent, month and tenure controls. Not a randomized trial.

Uncertainty

Table II, column 3: +0.301 resolutions/hour; agent-clustered standard error 0.0329; 12,295 agent-month observations. The text reports 15.2%, rounded to 15% in the abstract.

Quality

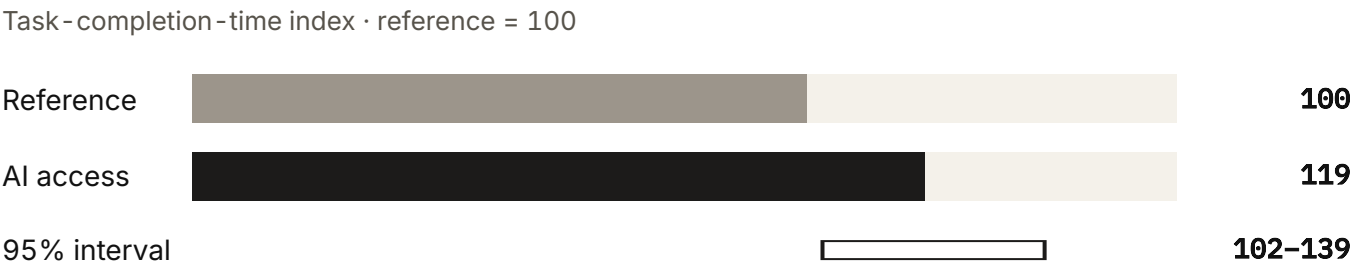
Effects vary by skill. The highest-skilled agents show small quality declines; resolution data cover a subset because subcontractor quality records are incomplete.

QJE: Generative AI at Work — source and limits

Experienced software developers

More time per completed task

Population: 16 experienced developers; 246 real tasks in mature open-source repositories they knew well.
Tools / study vintage: Primarily Cursor Pro with Claude 3.5 / 3.7 Sonnet; February–June 2025; methodology update 24 February 2026.



Transformation: $100 \times (1 + 19/100) = 119$. These are normalized reported relative estimates, not raw observed group means. Each panel has a different unit; the indices do not create a common productivity metric.

METR’s early-2025 experiment found 19% longer completion times in this selected setting; its February 2026 update warns that selection prevents a reliable estimate for later tools.

Historical interval: +2% to +39% task time (95%). The February 2026 update describes participation and task selection that prevent a reliable estimate for later tools.

Selected developers and familiar repositories, using early-2025 tools. Later participation and task selection changed: people and tasks expected to benefit increasingly avoided AI-disallowed assignment. This is not an estimate of current models.

Measure the time to finish the accepted task, including the work around generated output. A faster first draft and a slower completed task can coexist.

How to read this result

A plausible mechanism · source-supported account

Screen recordings and participant reports describe time spent prompting, checking and revising AI output. Such work can offset time saved producing a first attempt. The paper does not identify a causal percentage for each explanation.

The strongest alternative · analyst hypothesis

Experienced maintainers working in familiar repositories may have little to gain from these particular tools on these tasks. The selected population and early-2025 tools are an alternative to a general story about software work.

When a stronger model could matter · analyst hypothesis

More reliable edits, better repository context or a different workflow could change that balance. Neither this historical access experiment nor its comparison with customer support estimates the benefit of a stronger replacement model.

Design, uncertainty and quality

Comparison

Task-level random assignment: AI allowed versus AI disallowed

Design

Task-level randomized experiment after issues were specified. Log-linear estimation controls for forecasts elicited before random assignment.

Uncertainty

Historical estimate: +19% task time; 95% interval +2% to +39%, as restated in the February 2026 update. Self-reported time supported by screen recordings; some post-review time imputed.

Quality

No demonstrated qualitative reduction in submitted PR quality. Post-review time difference was statistically insignificant.

METR: original experiment and February 2026 update — sources and limits

These are two results to explain, not a verdict that customer support benefits while software does not. The narrower answer this edition can defend is that access changed completed work in these settings—and that interpreting the change requires following what workers actually did.

Where the work happens—and where the gain might go

The support case crosses organizational and national boundaries. The paper identifies these roles without naming the employer or assistant supplier.

Model developer	Assistant supplier	Employer / service provider	Workers	Customers
OpenAI developed the underlying GPT-3 model.	A separate, unnamed AI software company supplied the assistant integration.	An unnamed Fortune 500 business-software firm employed support agents directly and through subcontractors.	Most agents worked in the Philippines. The paper separately reports 89% outside the US; that is not a Philippine share.	Customers were US-based small-business owners seeking technical support.

An analytical attribution map. Equal-width arrows locate relationships in the service workflow; they encode neither effect sizes nor financial flows. Geography: final QJE paper §III.B and Table I.

Measured at the work: issues resolved per hour in this deployment. The estimated increase does not allocate the value among these actors.	Not measured by this result: wages, profits, fiscal gains, strategic leverage or welfare. Public institutions shape tax, labor and service rules; their fiscal gains, strategic leverage and welfare effects are not measured here.
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A future country comparison would need the location and scale of deployment, access terms, worker outcomes and value capture. The location of a model laboratory answers only one of those questions.

Brynjolfsson, Li & Raymond, QJE (2025); METR experiment (2025), with methodology update (24 February 2026). Draft use · author review pending

Evidence and caveats

Both exact uses await the finite author/release review. Primary contents checked 6 September 2026 by Codex; no original replication or named specialist review. [Full evidence trace](#)

Where does a faster task stop short of a finished outcome?

A design might arrive sooner while its test still takes just as long. Or AI could improve the test as well. The interaction makes both possibilities available—and separates completing one project from sustaining repeated output.

Scenario: change the workflow assumptions.

Illustrative arithmetic, not an empirical model. Change a complement, expand where AI helps, or alter the dependencies.

Hypothetical inputs throughout. Days and accepted units/week are invented teaching units. No input comes from a benchmark, S mark or country estimate. The two examples below are independent models.

One project: when is it finished?

Design takes 8 days; testing takes 32. Change the speed of either stage, or change how they depend on each other.

Added work is hypothetical and incremental to the baseline's existing work. It happens after the required tasks finish. Quality is held constant by assumption; no slider verifies it. Nothing is calibrated from METR.



Design 2 days · testing 32 days · added work 0 days
Scale: 0–80 days; fixed while speed, topology and added work change. Baseline edits change this scale.

40 → 34 days to complete one project.

6 days saved · 15% less elapsed time.

$$8/4 + 32/1 + 0 = 34 \text{ days}$$

Parallel means testing can proceed independently; it is a different process, not permission to test a design that does not yet exist. The optional added work is one lump after those tasks; queues, variable rework and quality changes remain unmodeled.

Repeated work: how much can keep coming out?

Three stations perform different steps on the same accepted unit. Dedicated stations can work on different units at once; the slowest capacity limits steady output.

Station 1		40/week
Station 2		4/week
Station 3		6/week

4 → 4 accepted units/week sustainable output ceiling.

$\min(40, 4, 6) = 4$ units/week

Binding station: 2.

Compatible units; adequate demand and buffers; no losses, rework, variability or start-up delay. Capacities are independently assumed, not inferred from the project durations. Project added work has no effect on these independently assumed capacities. All bars use a fixed 0–160 accepted-units/week scale.

In the default serial example, a design speedup saves six days before added work. Six extra days erase that saving; eight make the project take two days longer. The bottleneck in the station example follows from its assumed capacities. Neither example identifies an empirical bottleneck.

Hypothetical inputs adapted from the supplied author/editorial kit; equations and omissions shown at point of use. **SCENARIO**

The arithmetic locates a constraint only inside the assumptions you chose. Faster design helps this serial project until additional work absorbs the saving; parallel work changes the result again. In repeated production, raising the binding station's capacity can matter more than accelerating a station already ahead.

What to measure in a real workflow

A plausible real-world mechanism is that integrating or checking generated output adds work. An equally important alternative is that a better tool reduces those burdens or makes a different process viable. The illustration measures neither mechanism. It tells an investigator to record the whole completion path, quality threshold and station capacities before attributing an outcome to AI.

Which additional change would matter?

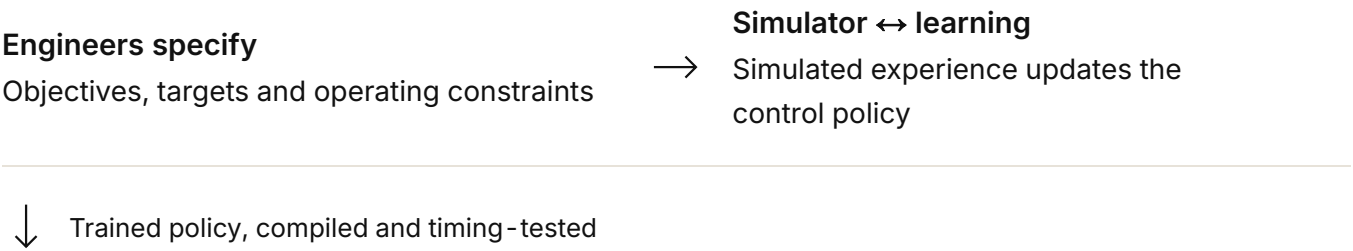
Fusion makes this question concrete. AI has already controlled plasma in physical hardware. The next question is what that result establishes, which conditions permit transfer, and how an improvement could reach an integrated plant outcome.

Observed: a real control result, with a longer evidence trail.

Training, physical deployment and evaluation in one research control workflow; wider plant requirements are separate dependencies.

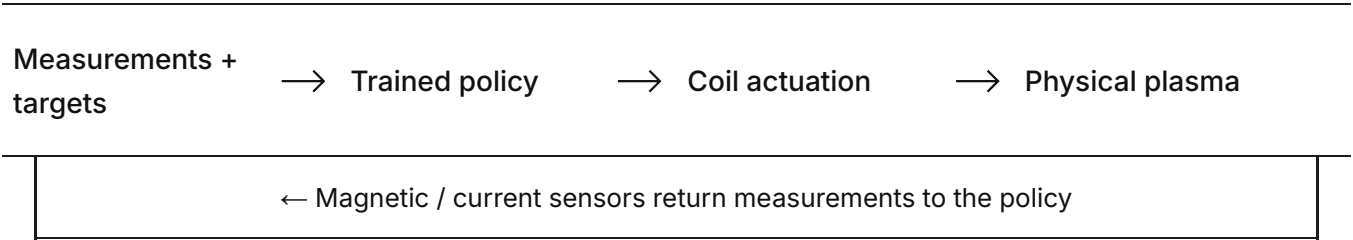
A specialist reinforcement-learning controller. This is a different kind of AI intervention from a general-purpose chat assistant. Its success does not measure the effect of upgrading today's chat models.

Train in simulation



Deploy the fixed policy on TCV

Existing control handles initial plasma formation before handover. The deployed policy's weights are not updated in this live loop.



Evaluate after the experiment

Reconstruct plasma states from measurements; compare performance with targets. This evaluation is outside the policy's live feedback loop.

Original explanatory schematic based on Degraeve et al., Figure 1 and Methods. Arrows show information/control relationships, not timing, safety certification or plant-level effect sizes.

Other plant dependencies

Plasma control does not cause materials qualification or fuel-cycle readiness. These are other requirements whose relationships need their own evidence.

Transfer to another setting: Test the controller across the operating conditions, sensing and actuation of the proposed next use. Simulator fidelity and hardware differences could change performance.

Materials and other plant systems: Seek material-specific exposure and post-irradiation evidence matched to an acceptance standard, alongside fuel-cycle and component tests. Planned irradiation is a requirement, not an accepted qualification dataset.

A useful plant outcome: Observe integrated performance, maintainability and availability over an appropriate period, with a defensible counterfactual for any attributed schedule or output gain. Evidence outside this bank could change the conclusion.

A simulation-trained reinforcement-learning policy controlled plasma configurations on the physical TCV research tokamak. Its training, deployment and evaluation required an engineered control workflow.

Choosing what the controller should achieve is substantive work. A system can meet a specified objective while leaving an important physical consideration outside that objective. Evaluation and revised specifications help define useful performance.

How to read this result

A plausible mechanism · source-supported account

The authors combined a learned policy with a physics simulator, hardware-specific sensing and actuation models, varied training conditions and real-time deployment engineering. The policy could control several plasma configurations on TCV.

The strongest alternative · analyst hypothesis

Success may depend on how well TCV and the tested conditions fit the simulator and engineering choices. This is an alternative to crediting general-purpose AI capability alone; the study does not isolate each contribution.

When a stronger model could matter · analyst hypothesis

The authors discuss richer policies and simulators as possible routes to broader performance. More capable learning could expand the feasible control task, while transfer and deployment requirements still need evidence. This specialist RL intervention does not measure chat-model upgrades.

Who defines useful control?

Extended Data Figure 4b describes an initially unspecified balancing objective: the controller used an alternative coil combination that produced forces on machine structures. Later experiments added an objective to address that behavior. The caption does not establish an accident.

Source observation: Extended Data Figure 4b. The governance inference is that objective specification and evaluation deserve scrutiny alongside the learned policy. This example is not evidence of a documented accident or existential risk.

Training

Engineers specify objectives and control targets. Simulation experience trains a control policy; only the trained policy is deployed to the live control loop.

Deployment

After initial plasma formation under existing control, the deployed policy reads magnetic/current measurements and targets, and sends commands to the coil power supplies.

Evaluation

Reconstructed plasma states support post-experiment assessment against targets; reconstruction is not the deployed policy's real-time shape observer. Compiled policies are timing-tested before deployment.

A research-control demonstration supports that task in its tested environment. Facility milestones do not establish AI causation; an absent result in this bank does not establish universal absence. AI might also improve other physical stages, which requires its own test. Degrave et al. — source, task and transfer limits · Article CC BY 4.0, subject to asset credit lines; diagram newly drawn, simplified and annotated.

Degrave et al., Magnetic control of tokamak plasmas through deep reinforcement learning, Nature (2022). Draft use · author review pending

Follow the gain to its beneficiary.

A stronger model could enable work that was previously unreliable or infeasible. It could also leave the next limiting step untouched, or introduce additional checking. To tell these possibilities apart, compare the actual intervention, the completed work and the people receiving the gain.

A stronger model could relax a quality threshold, improve a limiting stage, or make a different process viable. Another complement could be decisive instead. The contribution here is a way to make that question specific enough to investigate.

Draft personal copy · for Jinhua's final edit

I cannot infer a national dividend from the address of a model developer. In the support study, workers mainly in the Philippines serve US customers using an assistant built around an

American-developed model. I want to know where the work improved before deciding whose capacity improved.

Source references

Observation periods remain source-specific. These original sources support the figures; exact caveats and review states accompany the figures and the evidence page.

Survey on the Access to Finance of Enterprises in the euro area - Fourth quarter of 2025 · European Central Bank · 2026-02-02

https://www.ecb.europa.eu/stats/ecb_surveys/safe/html/ecb.safe202602.en.html

Use of artificial intelligence in enterprises · Eurostat · 2026-06-02

[https://ec.europa.eu/eurostat/statistics-explained/index.php?](https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Use_of_artificial_intelligence_in_enterprises)

[title=Use_of_artificial_intelligence_in_enterprises](https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Use_of_artificial_intelligence_in_enterprises)

The Microstructure of AI Diffusion: Evidence from Firms, Business Functions, and Worker Tasks · Kathryn Bonney; Cory Breaux; Emin Dinlersoz; Lucia Foster; John Haltiwanger; Aditya Pande · 2026-04-15

<https://www.census.gov/library/working-papers/2026/adrm/CES-WP-26-25.html>

Magnetic control of tokamak plasmas through deep reinforcement learning · Degraeve et al.; Nature · 2022-02-16

<https://doi.org/10.1038/s41586-021-04301-9>

Generative AI at Work · Brynjolfsson; Li; Raymond · 2025-02-04

<https://academic.oup.com/qje/article/140/2/889/7990658>

Measuring the Impact of Early - 2025 AI on Experienced Open - Source Developer Productivity (v2) · METR · 2025-07-25

<https://arxiv.org/html/2507.09089v2>

We are Changing our Developer Productivity Experiment Design · METR · 2026-02-24

<https://metr.org/blog/2026-02-24-uplift-update/>